

2.5 Assignment: Professional Portfolio Artifact 2

Chaitanya Chowdary Paladugu

Machine Learning Fundamentals (AIML-500)

Indiana Wesleyan University

Dr. Temilola Longe

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Portfolio Submission Links

Main Professional Portfolio: <https://chaitanyachowdarypaladugu.github.io/professional-portfolio/index.html>

Artifact 2 - Machine Learning vs. Deep Learning:

<https://chaitanyachowdarypaladugu.github.io/professional-portfolio/artifact-2.html>

Machine Learning vs. Deep Learning Report:

<https://chaitanyachowdarypaladugu.github.io/professional-portfolio/assets/artifact-2/>

[Chaitanya Machine Learning vs Deep Learning Portfolio.pdf](#)

Reflection

For my second professional portfolio artifact, I chose the Machine Learning vs. Deep Learning report that I created for Assignment 2.3. I designed the portfolio presentation for instructors, employers, and readers who want a clear example of how model selection depends on the problem and the available data. I used email spam detection and image recognition as familiar case studies and organized the page around the objective, traditional machine learning example, deep-learning example, direct comparison, real-world applications, skills, challenges, and value proposition. I also included links to the complete report and supporting documentation.

One important lesson I learned is that a more complex model is not automatically a better model. Logistic regression is a practical choice when useful features can be structured in advance and the task has a binary output. A convolutional neural network is more suitable when useful visual features must be learned directly from raw pixels. I also learned to distinguish between a documented product capability and an unverified claim about the exact model used inside a product. This helped me explain Gmail and Google Photos accurately while using logistic regression, CNN, and FaceNet research as appropriate technical examples.

I did not receive formal feedback from my instructor or classmates before completing this version. Instead, I used the assignment instructions, rubric criteria, source verification, and structured self-review. I checked that the comparison was clear, the citations and references were preserved in the detailed report, and the product claims were carefully qualified. I also tested the published portfolio page on desktop and mobile layouts and confirmed that the navigation, downloads, and final-report link worked correctly without horizontal overflow or browser-console errors.

Based on this review, I created a separate Artifact 2 page, added a clear project overview and objective, made the value proposition easy to find, and presented the logistic-regression and CNN comparison in a concise professional format. I listed Artifact 2 visibly in the portfolio, prioritized the final submission report, retained the detailed technical report as supporting evidence, and provided descriptive link labels. Overall, this project helped me understand how a professional portfolio can transform academic work into a focused demonstration of technical reasoning, model-selection knowledge, and clear communication.